

Jiajie Wu CV

Contact Information

Email: j.wu24@imperial.ac.uk

Education

PhD in Mathematics, Imperial College London

From 2024.09

M.S. in Mathematics, New York University

2022.09 - 2024.05

B.A. in Mathematics, New York University (with high honors)

2018.09 - 2022.05

Research Interests

Active Matter, Bio-fluid Dynamics, Cilia Dynamics, Collective Locomotion in Fluids, Delay Systems, Nonlinear Dynamics

Research Projects

Research at the NYU Applied Math Lab

2019.11-2024.06

Fluid-structure Interactions within Bird Flocks in Linear Formation Flights

Hydrodynamics of Finite-length Pipes at Intermediate Re

The Water Clocks

Research at Imperial College London

Since 2024.10

The Collective Dynamics of Cilia

Research at Helmholtz Pioneering Campus (Imperial-TUM Partnership)

Since 2024.11

The Physics of Mucus

Publications

C. Mavroyiakoumou, **J. Wu**, and L. Ristroph. "Modeling flying formations as flow-mediated matter," *Phys. Rev. Fluids* 11, 063103 (2026).

J. W. Newbolt, N. Lewis, M. Bleu, **J. Wu**, C. Mavroyiakoumou, S. Ramanarivo, and L. Ristroph. "Flow interactions lead to self-organized flight formations disrupted by self-amplifying waves," *Nat. Commun.* 15, 3462 (2024).

O. Pomerenk, S. Carrillo Segura, F. Cao, **J. Wu**, and L. Ristroph. "Hydrodynamics of finite-length pipes at Intermediate Reynolds Numbers," *J. of Fluid Mech.* (2023), vol. 959, A28.

Awards and Honors

Roth Research Training Support Grant (MATH_G00700)

2027-2028

Imperial-TUM Partnership Research Training Support Grant (MAAP_G44167)

2024-2027

Roth Scholarship

2024-2028

Graduate School of Arts & Science Tuition Scholarship

2022-2023

Phi Beta Kappa Honors Society

2022.04

Founders Day Award (University Honors Scholar)

2022.04

Larry & Klara Silverstein Research Scholar

2020.05

Dean's Undergraduate Research Fund

2020.05

Dean's List for Academic Year (all 4 academic years, 2018 - 2022)

Research Talks

The Physics of Mucus, Joint Academy of Doctoral Studies Summer School

2026.07

The Water Clocks, Dean's Undergraduate Research Conference

2021.05

Work Experience

Math & English Tutor, Shanghai Qingtu Education

Summer 2019

Course Grader, NYU Math Department

MA-UY 1124 Calculus II for Engineers

Fall 2020

MA-UY 4414 Applied Partial Differential Equations Fall 2020

Graduate Employee Adjunct Faculty, Courant Institute of Mathematical Sciences

MATH-UA 9 Algebra, Trigonometry & Functions Fall 2022

MATH-UA 264 Chaos and Dynamical Systems Spring 2023

MATH-UA 132 Math for Economics II Fall 2023

Graduate Teaching Assistant, Imperial College London

MATH50004 Multivariable Calculus and Differential Equations Fall 2024

MATH40007 Introduction to Applied Mathematics Spring 2025

MATH50003 Linear Algebra and Numerical Analysis Spring 2025

MATH50003 Linear Algebra and Numerical Analysis Fall 2025

MATH50003 Linear Algebra and Numerical Analysis Spring 2026

References

Dr. Eric Keaveny, Reader in Applied Mathematics, Imperial College London, e.keaveny@imperial.ac.uk

Prof. Leif Ristroph, Associate Professor of Mathematics, New York University, lr1090@nyu.edu

Prof. Michael Shelley, Lilian and George Lyttle Professor of Applied Mathematics, New York University and Director of Center of Computational Biology, Flatiron Institute, mshelley@flatironinstitute.org